

A Macro-Level Bandwidth Model for Monolithic 3D Memory

3d_memory — design-space exploration tool

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Abstract

We give the analytical kernel behind `3d_memory`, a macro-level design-space-exploration tool for monolithic 3D memories at the CACTI/DESTINY/NVSim altitude. Given a fixed storage capacity C , a fixed footprint A , a fixed layer budget L (each tier of physical thickness t_{layer}), and a peak power-density budget P_{max} (a cooling limit set by an external thermal stub), the tool searches 3D organizations that *maximize the internal array-supply bandwidth*. The central objects are a throughput-bound cycle time, a bandwidth identity linear in the number of independent peripheral sets, a volume-packing constraint that ties capacity, cell density, and the layer partition together, and a per-access energy model whose product with bandwidth is capped by P_{max} . Routing, the power-delivery network, and the thermal boundary condition that produces P_{max} are deferred (Sec. 6).

1 Problem statement

Given a fixed capacity C , a fixed footprint A , a fixed number of monolithic tiers L (each of physical thickness t_{layer}), and a peak power-density budget P_{max} (a cooling limit; the thermal stub of Sec. 6 maps ΔT_{max} onto it); *find* the 3D memory organization (array geometry, degree of peripheral sharing, sense-amp stacking, and tier assignment of circuit classes) that *maximizes* the internal array-supply bandwidth BW.

Two named 3D levers act on this objective:

1. **Peripheral stacking** — tucking sense amps, decoders, and selectors on tiers under/over the array (CUA / Choi-style), freeing footprint and relieving pitch.
2. **Multi-tier bitcells** — placing different devices of a single cell on different tiers, raising areal cell density.

2 Nomenclature

Symbol	Meaning
C, A, L	storage capacity, footprint (area/layer), layer budget
t_{layer}	physical thickness per tier (so $\text{VolBudget} = A L t_{\text{layer}}$)
ΔT_{max}	peak temperature rise allowed (thermal governor, feeds P_{max})
P_{max}	peak power-density budget [W/mm^2] (cooling limit)
a_{cell}	areal footprint of one stored bit (cell density $^{-1}$)
$N_{\text{arr},\text{indep}}$	number of <i>independent</i> peripheral sets (arrays or blocks of shared arrays)
$N_{\text{arr},\text{share}}$	number of arrays time-multiplexed onto one shared peripheral set
$N_{\text{arr},\text{tot}}$	total array count, $N_{\text{arr},\text{tot}} = N_{\text{arr},\text{indep}} \cdot N_{\text{arr},\text{share}}$
$N_{\text{SA}}, N_{\text{dec}}$	number of sense amps, number of decoders
b_{acc}	bits delivered per access (sense width)
$N_{\text{BL},\text{arr}}, N_{\text{WL},\text{arr}}$	bitlines, wordlines per array
t_{cycle}	steady-state cycle time (throughput bound)
$t_{\text{dec}}, t_{\text{WL}}, t_{\text{BL}}, t_{\text{SA}}$	decode, wordline, bitline-develop, and sense (incl. restore) times
t_{switch}	selector switching time for time-muxing shared arrays
$\text{Vol}_{\text{arr}}, \text{Vol}_{\text{SA}}, \text{Vol}_{\text{dec}}$	volumes of one array, sense amp, decoder
$\text{Vol}_{\text{sel}}, \text{Vol}_{\text{periph}}$	volumes of selector network, fixed periphery per array
$L_{\text{arr}}, L_{\text{periph}}$	layers holding bits vs. layers holding periphery
A_{SA}	footprint of one sense amp (constant)
K	column-multiplexing degree
$w_{\text{SA}}, p_{\text{BL}}$	sense-amp pitch, bitline pitch
u	areal utilization of a peripheral tier
$E_{\text{access}}, E_{\text{bit}}$	per-access energy, per-delivered-bit energy
P_{dyn}	dynamic power ($= \text{BW} \cdot E_{\text{bit}}$)
A_{used}	footprint the packed volume actually occupies
$N_{\text{cell},\text{tot}}$	total stored cells $= N_{\text{BL},\text{arr}} N_{\text{WL},\text{arr}} N_{\text{arr},\text{tot}}$

3 The kernel

3.1 Cycle time (throughput bound)

The steady-state cycle time is set by the slowest *un-hideable* resource occupancy. The shared decoder and sense-amp occupancies are floors that cannot be pipelined away; the array develop latency $t_{\text{WL}} + t_{\text{BL}}$ can be hidden by interleaving $N_{\text{arr},\text{share}}$ arrays onto the shared periphery:

$$t_{\text{cycle}} = \max \left(t_{\text{dec}}, t_{\text{SA}} + t_{\text{switch}}, \frac{t_{\text{dec}} + t_{\text{WL}} + t_{\text{BL}} + t_{\text{SA}} + t_{\text{switch}}}{N_{\text{arr},\text{share}}} \right). \quad (1)$$

t_{dec} is the critical decoder-stage delay; t_{SA} *includes restore* for destructive-read cells. This last inclusion silently encodes the technology dependence of sharing: for a destructive cell (DRAM) t_{SA} floors near t_{RC} and the shared term can never dominate, so interleaving buys nothing; for a non-destructive cell (gain cell, NVM) t_{SA} is small and interleaving pays.

3.2 Bandwidth

Internal array-supply bandwidth is the per-access word width divided by the cycle time, replicated across every *independent* peripheral set:

$$\boxed{\text{BW} = \frac{b_{\text{acc}}}{t_{\text{cycle}}} N_{\text{arr},\text{indep}}} \quad \text{with} \quad b_{\text{acc}} \leq N_{\text{BL},\text{arr}}. \quad (2)$$

The cap $b_{\text{acc}} \leq N_{\text{BL},\text{arr}}$ states that no access can sense more bits than there are bitlines (whole-row-activate limit). Sharing raises $N_{\text{arr},\text{tot}}$ and hides latency but never appears in (2): it adds capacity, not bandwidth.

3.3 Accounting identities

$$N_{\text{arr},\text{tot}} = N_{\text{arr},\text{indep}} \cdot N_{\text{arr},\text{share}}, \quad N_{\text{SA}} = N_{\text{arr},\text{indep}} \cdot b_{\text{acc}}, \quad N_{\text{dec}} = N_{\text{arr},\text{indep}}. \quad (3)$$

There is one decoder per independent set, and the sense-amp count is the total delivered word width.

3.4 Volume (packing identity)

Consumed volume equals the sum of every packed component — an equality, not a bound, so that no volume goes unaccounted:

$$\text{Vol} = \text{Vol}_{\text{arr}} \cdot N_{\text{arr},\text{tot}} + \text{Vol}_{\text{SA}} \cdot N_{\text{SA}} + \text{Vol}_{\text{dec}} \cdot N_{\text{dec}} + \text{Vol}_{\text{sel}}(N_{\text{arr},\text{share}}) + \text{Vol}_{\text{periph}} \cdot N_{\text{arr},\text{tot}}. \quad (4)$$

The selector term $\text{Vol}_{\text{sel}}(N_{\text{arr},\text{share}})$ is the volume of the network that time-multiplexes $N_{\text{arr},\text{share}}$ arrays onto one shared peripheral set; it grows with the sharing degree (concretely $\text{Vol}_{\text{sel}} \propto (N_{\text{arr},\text{share}} - 1) N_{\text{SA}}$, so it vanishes at $N_{\text{arr},\text{share}} = 1$) and is what makes sharing cost area, not just time. The final term $\text{Vol}_{\text{periph}} \cdot N_{\text{arr},\text{tot}}$ is a fixed per-array peripheral overhead (row decoders, wordline drivers, array edge). This same polynomial is reused as the occupied footprint $A_{\text{used}} = \text{Vol}/(L t_{\text{layer}})$ in the power-density constraint below.

3.5 Energy and power density

Each single-array read activates a whole $N_{\text{BL},\text{arr}}$ -wide row but delivers only b_{acc} bits (O'Connor row overfetch). The per-access dynamic energy is a uniform CV^2 form plus fixed periphery and an optional write term:

$$E_{\text{access}} = k_{\text{col}} N_{\text{BL},\text{arr}} + k_{\text{arr}} N_{\text{BL},\text{arr}} N_{\text{WL},\text{arr}} + e_{\text{periph}} + f_w e_{\text{write}} b_{\text{acc}}, \quad (5)$$

with $k_{\text{col}} = C_{\text{cell}} V^2$ (the per-bitline cell/access cap) and $k_{\text{arr}} = c_{\text{BL}} V^2$ (the activated row's distributed bitline cap, growing with rows). The swing is $V^2 = V_{\text{read}}^2$ for full-rail modes (SRAM latch, resistive-read precharge, and destructive charge-share write-back); the partial product $V_{\text{read}} V_{\text{sense}}$ applies only to a hypothetical *non-destructive* charge read. Amortizing over the delivered bits and multiplying by bandwidth gives the dynamic power, which together with a flat per-bit leakage/refresh term is held under the cooling budget as a *power density*:

$$E_{\text{bit}} = \frac{E_{\text{access}}}{b_{\text{acc}}}, \quad P_{\text{dyn}} = \text{BW} \cdot E_{\text{bit}}, \quad \frac{P_{\text{dyn}} + p_{\text{leak}} N_{\text{cell},\text{tot}}}{A_{\text{used}}} \leq P_{\text{max}}, \quad (6)$$

with $A_{\text{used}} = \text{Vol}/(L t_{\text{layer}})$ from (4). In the solver the density constraint is written division-free (moving A_{used} , itself the volume polynomial, to the right-hand side) so it stays polynomial for the spatial branch-and-bound. Because the footprint reuses the packed volume, energy, volume, and geometry are coupled through one expression; P_{max} is supplied by the external thermal stub of Sec. 6.

3.6 Scaling laws

$$t_{\text{WL}} = k_{\text{WL}}^{\text{wire}} N_{\text{BL, arr}}^2 + k_{\text{WL}}^{\text{cell}} N_{\text{BL, arr}}, \quad t_{\text{BL}} = f_{\Delta} (a_{\text{lin}} N_{\text{WL, arr}} + a_{\text{quad}} N_{\text{WL, arr}}^2), \quad (7)$$

$$\text{Vol}_{\text{dec}} \propto N_{\text{WL, arr}} \cdot N_{\text{BL, arr}} \propto \text{Vol}_{\text{arr}}, \quad \text{Vol}_{\text{SA}} = \text{const.} \quad (8)$$

The wordline develop t_{WL} keeps the wire/cell split: a distributed-line self-RC term $\frac{1}{2}rcL^2$ (CACTI-6 $n(n+1)rc/2$ form, quadratic in the $N_{\text{BL, arr}}$ cells along the line; coefficient $k_{\text{WL}}^{\text{wire}} = \frac{1}{2}rc$ set by the BEOL stack, *shared* across cell technologies) plus a linear gate-loading term $k_{\text{WL}}^{\text{cell}}$ (*per technology*).

The bitline develop t_{BL} is the **sense-signal settling time**, derived per sensing mode from Upton (2024) (current-mode readout, App. A; voltage-mode readout, App. B), with a charge-sharing branch for destructive DRAM/eDRAM cells. All three reduce to the same polynomial in the $N_{\text{WL, arr}}$ cells on the bitline: a linear *signal-development* term a_{lin} plus a quadratic BL wire self-RC term $a_{\text{quad}} = \frac{1}{2}r_{\text{BL}}c_{\text{BL}}$, scaled by a settling factor $f_{\Delta} = -\ln(1 - \Delta)$ that carries the sensing margin as a *time* cost. The settling fraction Δ (e.g. 0.99) is a fixed reliability requirement, not a free variable; tightening it costs time only logarithmically. The linear term is mode-specific:

- **current** (App. A, Eq. A.12): development time constant $\tau_C = C_{\text{BL}} n U_t / I_{\text{read}}$, so $a_{\text{lin}} = c_{\text{BL}} n U_t / I_{\text{read}}$.
- **voltage** (App. B, Eq. B.7): Elmore constant $\tau = v_{\text{ratio}}(R_{\text{pull-up}} + R_{\text{BL}}/2)C_{\text{BL}}$, pull-up-dominated, giving $a_{\text{lin}} = v_{\text{ratio}}R_{\text{pull-up}}c_{\text{BL}}$; the same divider ratio $v_{\text{ratio}} = V_{\text{BL, cell}}/V_{\text{read}}$ also multiplies the wire term a_{quad} . We use the single- τ settling latency $f_{\Delta} = -\ln(1 - \Delta)$, *not* the dual-edge clock-period bound $T_{\text{clk}} \geq -2\tau \ln(1 - \Delta)$ of Eq. B.8 — the $2\times$ is a half-cycle convention on the clock *period*, not on the serial develop latency, and dropping it keeps the shared wire term consistent across modes.
- **charge_share** (DRAM/eDRAM, DESTINY convention): passive charge redistribution with a lumped series-RC $a_{\text{lin}} = r_{\text{BL}}C_{\text{cell}}$ (CACTI-D / CACTI-5.1) plus the same distributed wire self-RC a_{quad} , which overtakes the lumped term once $N_{\text{WL, arr}} \gtrsim 2C_{\text{cell}}/c_{\text{BL}}$ — the long, thin, resistive 3D-BEOL bitline regime this tool targets. The series-cap ratio governs signal *amplitude*, not speed, so it enters only as the row cap of Sec. 4.

t_{SA} retains the intrinsic device read/latch floor (and restore, for destructive cells). Decoder volume tracks the array it drives; sense-amp volume is a fixed footprint per amp — the earlier Pelgrom area law $A_{\text{SA}} \propto 1/\text{margin}^2$ is dropped, because the accurate cost of a sensing margin is the settling *time* above, not sense-amp area.

4 Derived structure

4.1 Layer partition

Capacity and footprint fix how many tiers must hold bits; the rest are free for periphery:

$$L_{\text{arr}} = \left\lceil \frac{C a_{\text{cell}}}{A} \right\rceil, \quad L_{\text{periph}} = L - L_{\text{arr}}, \quad N_{\text{SA}} \leq \frac{A L_{\text{periph}} u}{A_{\text{SA}}}. \quad (9)$$

Bandwidth lives in L_{periph} . Hence **cell density is a quantified indirect bandwidth lever**: shrinking a_{cell} (e.g. by making the cell 3D / multi-tier) lowers L_{arr} and frees peripheral tiers for more sense amps — worthwhile iff the freed tiers buy more N_{SA} than the added cell complexity costs in Vol_{arr} .

4.2 Charge-share row cap

For a destructive charge-sharing cell the developed signal amplitude $C_{\text{cell}}/(C_{\text{cell}} + c_{\text{BL}} N_{\text{WL, arr}})$ must stay above the sense-amp offset margin m_{SA} . This is an amplitude, not a speed, constraint, so it enters as a constant upper bound on the rows rather than a develop-time term:

$$N_{\text{WL, arr}} \leq \frac{C_{\text{cell}} (1 - m_{\text{SA}})}{c_{\text{BL}} m_{\text{SA}}}. \quad (10)$$

This is why DESTINY caps DRAM subarray rows; for non-destructive (current/voltage) cells the bound is absent.

4.3 Optimal sharing degree

Interleaving helps only until the develop term in (1) drops to the floor; beyond that, more sharing adds Vol_{sel} and $N_{\text{arr, tot}}$ for no bandwidth:

$$N_{\text{arr, share}}^* = \left\lceil \frac{t_{\text{dec}} + t_{\text{WL}} + t_{\text{BL}} + t_{\text{SA}} + t_{\text{switch}}}{\max(t_{\text{dec}}, t_{\text{SA}} + t_{\text{switch}})} \right\rceil. \quad (11)$$

4.4 Width / mux / pitch coupling

The sense width is set by the column-mux degree, which is floored by the sense-amp pitch; stacking n sense-amp tiers relieves that pitch:

$$b_{\text{acc}} = \frac{N_{\text{BL, arr}}}{K}, \quad K \geq \left\lceil \frac{w_{\text{SA}}}{p_{\text{BL}}} \right\rceil / n. \quad (12)$$

So the width lever, the mux degree, the sense-amp pitch, and sense-amp tier stacking are a single coupled knob.

5 Allocation recipe

Given $(C, A, L, \text{cell/tier tech})$, the optimizer walks the levers in dependency order:

1. **Cell density** — fixes L_{arr} via (9), freeing L_{periph} .
2. **Array geometry** — fixes the t_{cycle} floor (1) and $\text{Vol}_{\text{arr}}, A_{\text{SA}}$.
3. **Sense-amp stacking + mux degree** — raises b_{acc} toward $N_{\text{BL, arr}}$ (12).
4. **Share to $N_{\text{arr, share}}^*$** — reaches the t_{cycle} floor at minimum cost (11).
5. **Fill L_{periph}** — spend remaining peripheral area on N_{SA} (9), maximizing (2), until either the peripheral area or the power-density budget (6) binds.

6 Scope and deferrals

The model is a *packing-plus-power* model: how much bandwidth-producing hardware fits in L tiers of area A at capacity C without exceeding the cooling power density $P_{\max}$ (6). Two assumptions justify deferring routing: (i) **distributed vertical bottom-exit** — each independent array exits its bits through a local per-array vertical port grid, so there is no die-scale horizontal data-collection network; and (ii) **consumer-provisioned** — we model that the bandwidth exists at the exit surface, not who consumes it. Under these, data egress collapses to a short local vertical hop (roughly uniform across banking choices), so the packing result is close to physical rather than a loose upper bound. The internal selector cost is retained (t_{switch} , Vol_{sel}).

Deferred: the thermal boundary condition that *sets* $P_{\max}$ (a pluggable stub maps an areal R_{th} and $\Delta T_{\max}$ to the power-density budget consumed in (6)); the power-delivery network; and the lateral address/command distribution (narrow, assumed provisioned). Note the power-density *constraint* itself is no longer deferred — it is enforced directly through the per-access energy model of Sec. 3.